Jessie Oehrlein

Assistant Professor of Mathematics · Fitchburg State University

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Education

Columbia University *New York City, NY*
PhD Applied Mathematics - Atmospheric Science *8/2016 - 5/2021*
- Dissertation: *Sudden Stratospheric Warmings and Their Impact on Northern Hemisphere Winter Climate*
- Advisor: Dr. Lorenzo Polvani
- MS Applied Mathematics 5/2017
- MPhil Applied Mathematics 5/2019

Franklin W. Olin College of Engineering *Needham, MA*
BS Mechanical Engineering *8/2012 - 5/2016*
- Capstone: *Mathematical Modeling of Braided Gastrointestinal Stents* sponsored by Boston Scientific

Professional Experience

2021-	**Assistant Professor of Mathematics**, Mathematics Deparatment, Fitchburg State University
2020	**Adjunct Lecturer**, Mathematics, Engineering, and Computer Science, LaGuardia Community College
2019-20	**Graduate Teaching Assistant**, Applied Physics & Applied Mathematics, Columbia University
2012-2020	**Grader & Teaching Assistant**, Art of Problem Solving
2019	**Instructor**, Columbia University Summer Bridge Program
2016	**Engineering Intern**, Boston Scientific Corporation

Teaching Experience

Fall 2021 - present	**Fitchburg State University**, MATH 1300 Functions SI, MATH 1600 Informal Mathematical Modeling, MATH 1700 Applied Statistics, MATH 1900 Discrete Mathematics, DATA 2000 Principles of Data Analysis, MATH 2800 Statistical Analysis, MATH 3003 Advanced Statistics, MATH 3350 Multivariate Calculus, MATH 3900 Mathematics Seminar (Sp23: time series, Sp26: mathematics of climate), MATH 4200 Probability & Statistics I, MATH 4250 Probability & Statistics II
Summer 2022 & 2026	**MathPath**, Summer math camp for ages 11-14. Climate & Chaos, The Amazing Algorithmic Race, Graph Coloring, Conditional Probability and Bayes' Theorem
Summer 2026	**National Math Camp**, Summer math camp for ages 11-14. The Math of How Things Change (Dynamical Systems), The Art of Algorithms
2020	**LaGuardia Community College**, MAC 281 Discrete Structures, MAC 286 Data Structures
2019 - 2020	**Columbia University**, Introductory Statistics (Bridge Program Instructor), APMA E3001 Multivariable Calculus for Engineers and Applied Scientists (Graduate TA)
2012 - 2020	**Art of Problem Solving**, Introductory and intermediate mathematics (prealgebra, algebra, geometry, number theory, counting & probability), programming (Python, Java), and math contest preparation courses for K-12 students (grader & TA)

Publications

Published Research

Note: Authorship in mathematics is typically by last name. Authorship in other fields here is typically by contribution.

J.J. Sylvia IV, Z. Miner, **J. Oehrlein**. 2025. "Impacts of Intersectional Dual-Perspective VR Experiences on Empathy and Understanding of Microaggressions." *Human Technology*, 21(1), 73-105. doi:10.14254/1795-6889.2025.21-1.4.

Y. Yesilevskiy, A. Thomas, **J. Oehrlein**, M. Wright, M. Tarnow. 2022. "Introducing Experimental Design to Promote Active Learning." American Society for Engineering Education Annual Conference & Exposition 2022 Proceedings.

J. Oehrlein, L. M. Polvani, L. Sun, C. Deser. 2021. "How well do we know the surface impact of sudden stratospheric warmings?" *Geophysical Review Letters*. doi: 10.1029/2021GL095493.

J. Oehrlein, G. Chiodo, L. M. Polvani. 2020. "The effect of interactive ozone chemistry on weak and strong stratospheric polar vortex events." *Atmos. Chem. Phys.* . doi: 10.5194/acp-20-10531-2020.

J. Oehrlein, G. Chiodo, L. M. Polvani. 2019. "Separating and quantifying the distinct impacts of El Niño and stratospheric sudden warmings on North Atlantic and Eurasian wintertime climate variability." *Atmospheric Science Letters*, 20 (7), e923.

G. Chiodo, **J. Oehrlein**, L. M. Polvani, J. C. Fyfe, A. K. Smith. 2019. "Insignificant influence of the 11-year solar cycle on the North Atlantic Oscillation." *Nature Geoscience*. doi: 10.1038/s41561-018-0293-3.

M. Beaudouin-Lafon, S. Chen, N. Karst, **J. Oehrlein**, D. S. Troxell. 2017. "Labeling Crossed Prisms with a Condition at Distance Two." *Involve*, 11(1), p. 67-80.

N. Karst, J. Langowitz, **J. Oehrlein**, D. S. Troxell. 2017. "Radio k-Chromatic Number of Cycles for Large k." *Discrete Mathematics, Algorithms, and Applications*, 9(3).

N. Karst, **J. Oehrlein**, D. S. Troxell, J. Zhu. 2015. "L(d,1)-labelings of the edge-path-replacement by factorization of graphs." *J. of Comb. Optimization*, 30 (1), p. 34-41.

N. Karst, **J. Oehrlein**, D. S. Troxell, J. Zhu. 2015. "On distance labelings of amalgamations and injective labelings of general graphs." *Involve*, 8 (3), p. 535-540.

N. Karst, **J. Oehrlein**, D. S. Troxell, J. Zhu. 2015. "The minimum span of L(2,1)-labelings of generalized flowers." *Discrete Applied Math.*, 181, p. 139-151.

N. Karst, **J. Oehrlein**, D. S. Troxell, J. Zhu. 2014. "Labeling amalgamations of Cartesian products of complete graphs with a condition at distance two." *Discrete Applied Math.*, 178, p. 101-108.

Published & Accepted Teaching Materials

J. Oehrlein, 2025. "Normal Distributions." *POGIL Activity Clearinghouse*, 6(1).

J. Oehrlein, 2025. "Percentiles & Boxplots." *POGIL Activity Clearinghouse*, 6(1).

J. Oehrlein, 2025. "Data Basics." *POGIL Activity Clearinghouse*, 6(1).

J. Oehrlein, 2023. "Confidence Interval Interpretation." *POGIL Activity Clearinghouse*, 4(1).

J. Oehrlein, 2022. "Using Project EDDIE Modules in Applied Statistics." Adapted & edited materials and instructor story, Carleton SERC Project EDDIE.

J. Oehrlein, to appear. "Sampling Methods." *POGIL Activity Clearinghouse*.

J. Oehrlein, to appear. "Wrangling Rows and Columns." *POGIL Activity Clearinghouse*.

In Review

J.L. Talanian, B. Mgeni, **J. Oehrlein**. "Effects of Including Rehearsal Push-Ups to a Dynamic Warm-Up Prior to Completing a Maximal Push-Up Test."

Awards & Fellowships

2025, 2026 **MSCA Professional Development Funds**, Massachusetts State College Association Union

2024 **IBL Travel Grant**, Inquiry-Based Learning Special Interest Group of the Mathematical Association of America

2021-2022 **Project NExT Fellow**, Mathematical Association of America

2019-2021 **Teaching Assessment Fellow**, Columbia Center for Teaching & Learning

2019 **Outstanding Student Presentation Award**, American Meteorological Society Conference on Middle Atmosphere

2016-2021 **Graduate Research Fellow**, National Science Foundation

Presentations

Invited Talks & Guest Lectures

June 2026. *Discovering the Process Along the Way*, Electronic Conference on Teaching Statistics Closing Session Lightning Talk.

May 2025. K. Tracy, **J. Oehrlein**. *Engaging Students in Coursework During Their First Year*, Fitchburg State University Center for Teaching & Learning Summer Institute: Using Pedagogy and Assessment to Foster Student Engagement, Fitchburg, MA.

Dec. 2024. C. Oehrlein, **J. Oehrlein**. *Creating Models in the Classroom*, Process-Oriented Guided Inquiry Learning (POGIL) eSeries.

May 2024. Panel on objectivity. HON 1200 Honors English II, Fitchburg State University, Fitchburg, MA.

Mar. 2024. Statistics follow-up to a talk by Dr. Elise Takehana. MATH 1850 Freshman Seminar in Mathematics, Fitchburg State University, Fitchburg, MA.

Springs 2023-26. *Math in the Sky: From Winter Weather to Future Climate*. Math 1850 Freshman Seminar in Mathematics, Fitchburg State University, Fitchburg, MA.

Feb. 2023. *Data Representation*. Guest Lecture in CSCI 327 Inclusive Technology Design, Oberlin College, Oberlin, OH.

Dec. 2022. *Math in the Sky: From Winter Weather to Future Climate*. Olin College Research Seminar, Needham, MA.

Oct. 2022. *Understanding the Atmosphere: From Winter Weather to Future Climate*. Olin College Alumni Speaker Series, Needham, MA.

C. Evans, **J. Oehrlein**. Aug. 2022. *Your First Years as Faculty*. Preparing to Teach Statistics and Data Science workshop, Fairfax, VA.

Oct. 2021. *Math in the Sky: Studying the Stratospheric Polar Vortex*. Joint Mathematics & Geology, Environment and Sustainability Seminar, Hofstra University, Hempstead, NY.

Sept. 2020. *Using Climate Models to Understand Stratosphere-Troposphere Interaction.* Talk Math With Your Friends Virtual Colloquium.

Aug. 2020. *The Other Polar Vortex: Stratospheric Impacts on North Atlantic Winter.* SURF at DAWN REU Summer Lunch Talk, University of Copenhagen, Denmark.

Selected Contributed Presentations

Aug. 2026. *Teaching with Strands in Introductory Statistics*. MAA MathFest. Contributed Poster Session. Boston, MA.

Jun. 2026. *Teaching with Strands in Introductory Statistics*. Electronic Conference on Teaching Statistics, Posters & Beyond, Virtual. Jan. 2026. *Bringing the Stratosphere to Statistics*. American Meteorological Society Conference on Education & Conference on Middle Atmosphere. Session on Beyond the Basics: Innovative Teaching Strategies for Advanced Atmospheric Topics. Houston, TX.

Aug. 2025. *Writing Statistics and Data Science Activities with Students' Process in Mind*. MAA MathFest. Contributed Paper Session on Inquiry Based Learning. Sacramento, CA.

C. Oehrlein, **J. Oehrlein**. Jul. 2025. *Exploring models, building collaboration: Intro to Process-Oriented Guided Inquiry Learning in statistics*. US Conference on Teaching Statistics Breakout Session, Ames, IA.

Jun. 2025. *Building a Process Skill Sequence Through a Course*. National Conference on Advancing POGIL Practice Roundtable, Atlanta, GA.

Jun. 2025. *Making Summative Assessment a Little More POGIL-y*. National Conference on Advancing POGIL Practice Poster Session, Atlanta, GA.

Nov. 2024. *Focusing on Interpretation and Evaluation in Statistics & Data Science*. American Mathematical Association for Two Year Colleges Annual Conference mini-session, Atlanta, GA.

Aug. 2024. *Losing the Model in the Calculations: Classifying Critical Points*. MAA MathFest, Contributed Paper Session on Teaching Flops: Learning and Adapting When Teaching Goes Astray, Indianapolis, Indiana.

Jun. 2024. *Inquiry-Based Learning in an Undergraduate Probability Theory Course*. Electronic Conference on Teaching Statistics, Posters & Beyond, Virtual.

J. L. Talanian (presenter), B. Mgeni, **J. Oehrlein**. May 2024. *Completing the ACM Push-up Test Following a Dynamic Warm-up With and Without Rehearsal Push-ups*, American College of Sports Medicine Annual Meeting Poster Session.

W. Flynn (presenter), R. Tanamachi, **J. Oehrlein**, Z. J. Handlos, R. H. Humphrey, S. C. Petitto. Jan. 2024. *Thinking About Thinking: How is Metacognitive Development Fostered in Undergraduate Atmospheric Science Programs?*. American Meteorological Society 33rd Conference on Education, Atmospheric Science Education Research II.

Aug. 2023. *Reading & Critiquing Applications of Statistics in an Introductory Course.* MAA MathFest, Contributed Paper Session on My Favorite Statistics/Data Science Activity, Tampa, FL.

A. Theobold, C. Evans, **J. Oehrlein**, S. Stoudt. Jun. 2023. *Alternative grading: a more meaningful representation of student learning*. US Conference on Teaching Statistics Breakout Session, University Park, PA.

Jan. 2023. *Integrating Modeling Into a Functions Corequisite.* Joint Mathematics Meetings, COMAP Session on Integrating Modeling into Established Courses, Boston, MA.

K. Spoon, L. Feinman, **J. Oehrlein**, C. Holdines. Nov. 2022. *Alternative Grading Practices Panel*. American Mathematical Association for Two Year Colleges Annual Conference, Toronto, ON.

Jan. 2021. *Mathematics Research-Like Experience in a Discrete Structures Course for Computer Science Students.* Joint Mathematics Meetings, MAA Session on Inquiry-Based Learning and Teaching, Virtual.

Jan. 2021. *Teaching Data Literacy Through a Stratosphere Class for Secondary Students.* American Meteorological Society 30th Conference on Education Poster Session, Virtual.

Jan. 2021. *Teaching Disciplinary Practices in Atmospheric Science and Discrete Mathematics*. Applied Physics & Applied Mathematics Research Seminar, Columbia University, New York, NY.

J. Oehrlein, L. Sun, L.M. Polvani, C. Deser. Dec. 2020. *Characterizing the Surface Impact of Sudden Stratospheric Warmings in the Context of Internal Variability.* American Geophysical Union Fall Meeting, Subseasonal to Seasonal Climate Prediction, Processes, and Services, Virtual.

J. Oehrlein, L. Sun, L.M. Polvani, C. Deser. Aug. 2020. *Bootstrap-Based Identification of the Surface Impact of Sudden Stratospheric Warmings.*, Joint Statistical Meetings, Contributed Session on Climate and Meteorological Statistics, Virtual.

Jan. 2020. *Maps, Bridges, Networks, and Art Galleries: Introducing Secondary Students to Graph Theory through Classic Problems*. Joint Math Meetings, MAA Session on Outreach, Teaching and Learning Advanced Mathematics, Denver, CO.

J. Oehrlein, G. Chiodo, L.M. Polvani. Jan. 2019. *Separating and quantifying the distinct impacts of El Niño and stratospheric sudden warmings on North Atlantic and Eurasian wintertime climate.* American Meteorological Society 20th Conference on Middle Atmosphere, Role of the Stratosphere in Climate Variability, Change, and Prediction, Phoenix, AZ.

Sessions Organized

Jan. 2026. Beyond the Basics: Innovative Teaching Strategies for Advanced Atmospheric Topics. American Meteorological Society Conference on Education & Conference on Middle Atmosphere, Houston, TX. Co-organized with Staci DeSchryver.

Aug. 2025. Mini-Course on Introduction to Process Oriented Guided Inquiry Learning (POGIL) for Math and Stats Courses. MAA MathFest, Sacramento, CA. Co-organized with Chris Oehrlein and Kayla Heffernan.

Aug. 2025. Contributed Paper Session on Inquiry Based Learning. MAA Mathfest, Sacramento, CA. Co-organized with Lee Roberson, Joe Barrera, Mel Henriksen, Mami Wentworth, Rebekah Jones, Chris Oehrlein, Katie Johnson, Kayla Heffernan.

Jun. 2023. Birds of a Feather: Reinforcing or Reintroducing Pre-req Skills. National Conference on Advancing POGIL Practice, Salt Lake City, UT.

Jun. 2023. Birds of a Feather: Teaching Corequisites. US Conference on Teaching Statistics, University Park, PA.

Jul. 2022. So You Want to Build a New Course. MathFest Project NExT Discussion.

Mar. 2022. Establishing Interdisciplinary Collaborations in Teaching and Research. Joint Math Meetings Project NExT Session. Co-organized with Jamie Haddock.

Research Mentoring

2025-2026	**Aidan Thompson**, Independent Study Advisor, Fitchburg State University
2023-2024	**Cooper Rogers**, Honors Advisor, *Creating a Study Guide for the Actuarial P Exam*. Fitchburg State University
2023-2024	**Benjamin Wiita**, Honors Advisor, *Are MLB MVP Voters Getting Better?*, Fitchburg State University
2024	**Samantha Chandler**, Honors Thesis Second Reader, *Artist's Companion for Perspective Drawing*, Fitchburg State University
2019-2021	**Johanna Doyle**, Research Advisor, *El Niño-Southern Oscillation and Final Stratospheric Warmings*, Bronx High School of Science

Professional Service & Development

Internal Service

2024 -	**Pi Mu Epsilon Mathematics Honors Society**, Faculty Sponsor
2022 - 2026	**Mathematics Department Assessment Committee**, Committee Member, Chair 2024-26
2022 - 2026	**Mathematics Department Haskins Contest**, Contributor, Editor, Solutions Session
2023 - 2026	**American Mathematics Competition 10/12 Host Site**, Organizer, Proctor
Fall 2025	**Search Committee: Dean of the Library**, Member
Sum 2025	**Mathematics Supplemental Instruction Analysis**, Data analyst, work supported by SU-SUCCESS funds
2023 - 2025	**Center for Teaching & Learning Council**, Member, Grant reviewer
2024 - 2025	**University Assessment and Research Committee**, Member, Peer reviewer
Spring 2025	**AUC Curriculum Committee**, Member, Rotating secretary
2021 - 2024	**AUC Academic Policies**, Committee Member, Co-Chair 2023-24
2021 - 2025	**Mathematics Department Curriculum Committee**, Committee Member, Chair 2022-23 and Spring 2024
Fall 2024	**Mathematics Department TT Hiring Committee**, Chair
Sp 2024	**Mathematics Department VAP Hiring Committee**, Committee Member
Sum 2023	**Mathematics Placement Analysis**, Data analyst, work supported by Dean of Health & Natural Sciences
2021 - 2023	**Ad-hoc Data Science/Analytics Committee**, Developed new Data Analytics minor
2022 - 2023	**Leading for Change Climate Survey Dissemination Committee**, Member
Fall 2022	**Fall Open House**, Department Representative
2021 - 2022	**Crocker Center Advisory Board**, Member

External Involvement, Service, and Outreach

2023 - 2029	**Mathematical Association of America Membership Committee**, Committee Member, chair 2026-29, co-organized two-year college roundtable
2025-2028	**American Meteorological Society Committee on Middle Atmosphere**, Outreach Committee Member, co-organized 2026 biennial conference
2024-	**POGIL Activity Clearinghouse Content Team**, Serve as curator for activities post-review
2023 - 2024	**Advanced Placement Statistics**, Exam Reader (scoring)
2021 - 2026	**Undergraduate Statistics Project and Data Scrollytelling Competitions**, Student Paper Judge
2022 - 2025	**Skype a Scientist**, Spoke to Girl Scout Troop in NY, Math Circle in MN, Earth Science class in NJ, 2nd Grade class in TN
2020 - 2024	**COMAP Mathematical Contest in Modeling**, Student Paper Judge
2018 - 2024	**Association for Women in Mathematics Student Essay Contest**, Student Paper Judge
2023 - 2024	**MAA Mathfest**, Student Poster Presentation Judge
2018-2020, 2023	**Joint Mathematics Meetings**, Student Poster Presentation Judge
2024	**American Meteorological Society Conference on Middle Atmosphere**, Student Presentation Judge
2024-26	**UConn Sports Analytics Symposium Data Challenge**, Student Paper Judge
2023	**Math Prize for Girls**, Alumnae Panel Member
2023	**NSF Panel**, Reviewer / Panel Member
2022	**SIMIODE Challenge Using Differential Equations Modeling**, Student Paper Judge
2022	**Regeneron International Science & Engineering Fair American Statistical Association special award**, Student Paper Judge

Development

Fitchburg State Center for Teaching & Learning Events, Fall 2021-present. Attended multiple Teaching & Learning hour events and summer institutes.

Process-Oriented Guided Inquiry Learning Writers' Retreat, Summers 2022 & 2024. Worked with fellow authors and coaches to develop POGIL-style activities.

Spring 2024 Statistics Education Faculty Learning Community, met biweekly through Spring 2024 to develop, revise, and refine statistics and data science classroom activities.

Early Career Statistics Faculty Group, Fall 2022- present. Meet monthly to share resources and support.

Developing Expertise and Building Collaborations to Advance Atmospheric Science Education Research, Summer 2023, NSF-funded workshop to build research groups and capacity in atmospheric science education research.

Process-Oriented Guided Inquiry Learning (POGIL) Learning Community. 2021 - 2023, writing-focused in year 2. Met monthly to discuss POGIL activity authoring and facilitation.

Project NExT. 2021 - 2022, early career math faculty development through Mathematics Association of America.

National Association of Geoscience Teachers Early Career Workshop. Summer 2022, development workshop for early career faculty across earth and space sciences.

Peer Review

Journal of Climate
Geophysical Research Letters
npj Climate and Atmospheric Science
Journal of Geophysical Research: Atmospheres
Atmosfera
MDPI Remote Sensing

MDPI Meteorology
Teaching & Learning Inquiry
Process-Oriented Guided Inquiry Learning (POGIL) Activity Clearinghouse
POGIL Collection Endorsement (2022, 2024)